

## Gemini Hierarchical Line Drawing Prompt Appendix

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The goal of the workflow is to convert an architectural photo into a hierarchical set of black-and-white line drawings with **Nano Banana 2** (`gemini-3.1-flash-image-preview`). The current implementation has two parts:

1. **Stage 0-3**: generate one **2×2 grid** in a single Gemini call;
2. **Stage 4-6**: generate one refined line drawing per stage, using the original image and the previous stage as inputs.

The stage vocabulary used in this document is: **Stage 0**: building complex / building; **Stage 1**: section; **Stage 2**: floor; **Stage 3**: windows / doors; **Stage 4**: structural refinement; **Stage 5**: simple ornament; **Stage 6**: full detail

### 1. Inputs and generation settings

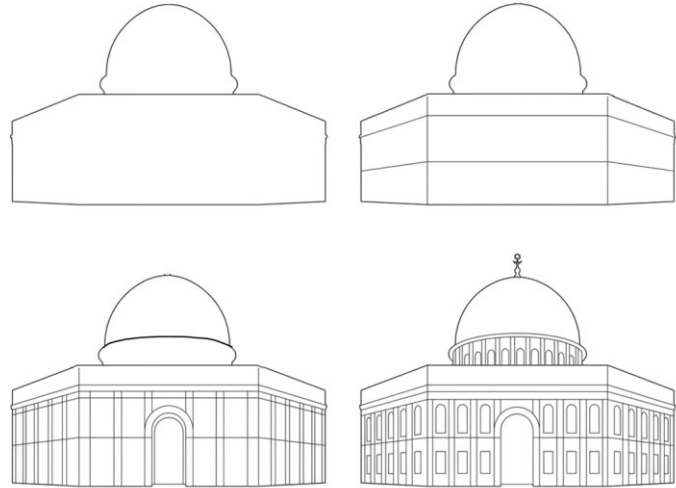
#### 1.1 Stage 0-3 grid generation

Gemini receives three inputs:

1. the original building image `original_input.png`;
2. the example grid image `static/Example.png`, used only as a reference for layout, abstraction level, and stage progression;
3. the **Stage 0-3 grid prompt**.



Original image



Reference layout

Figure 1: the original building image and the example grid image

#### 1.2 Stage 4-6 refinement generation

Gemini receives three inputs:

1. the original building image `original_input.png`;
2. the previously generated stage image;
3. the refinement prompt template, with the current stage name, description, and stage-specific instruction inserted at runtime.

## 2. Stage 0-3: unified 2×2 grid prompt

In the current implementation, Stage 0-3 are **not** generated as four separate Gemini calls. They are generated together in a single 2×2 grid prompt.

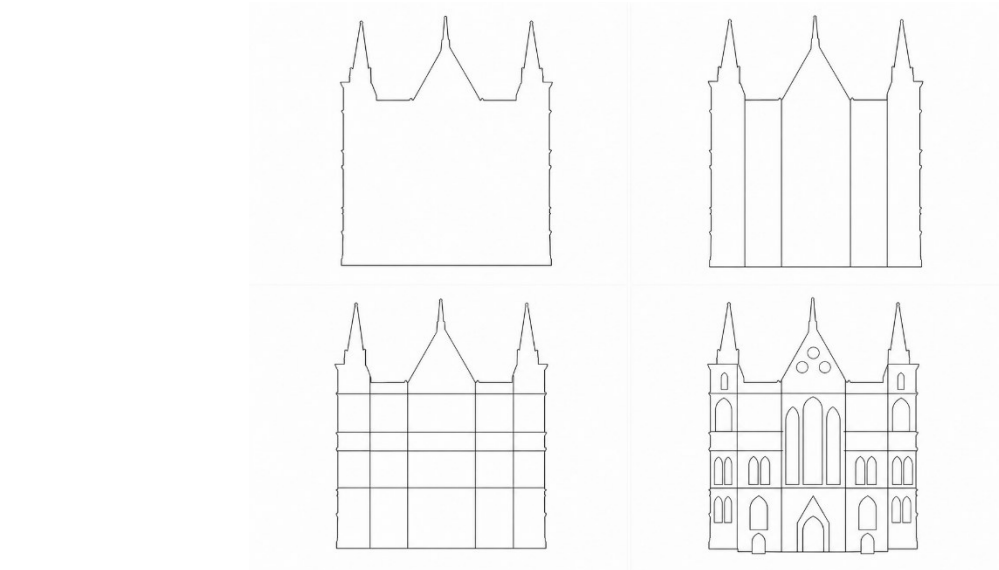


Figure 2: the result of stage 0-3

### Prompt:

You are an architectural hierarchy renderer for the project

Goal:

Transform the input building photo into exactly one 2x2 grid image that contains stages 0 to 3 of the same building.

Reference images:

- The first reference image is the original building photo and is the only source for the building's geometry, proportions, viewpoint, and visible architectural content.
- The second reference image is an example hierarchy grid for reference only.
- Use the example image only to understand the intended 2x2 layout, stage progression, abstraction level, and visual style.
- Do not copy, trace, or reuse the building shape, composition, or details from the example image.
- The output must still depict only the building from the original photo.

Output format:

- Output one image only.
- The output must be a strict 2x2 four-panel grid.
- Each quadrant must occupy exactly one quarter of the canvas.
- No captions, no labels, no numbers, no arrows, no legends, and no decorative borders.
- All four quadrants must keep the same building viewpoint, framing, scale, perspective, and proportions.
- No line, stroke, filled region, or building fragment may cross a quadrant boundary.
- Do not place any extra geometry or fragments in the center horizontal or vertical gutter.
- Any area that is not a thin line drawing must remain pure white.

Global rules:

- Preserve the original building's proportions, silhouette, framing, perspective, and overall geometry. - Focus only on the main building or main building group. Ignore all secondary, background, marginal, or unrelated structures. - For stages 0 to 3, use only the simplest geometric forms and a minimal number of single clean lines. - Reduce the building to simple masses, boundaries, and openings. Keep the representation abstract and diagrammatic. - Do not draw facade-specific fine articulation in stages 0 to 3. - Keep only the main building subjects. Remove sky, clouds, trees, ground textures, street furniture, people, cars, shadows, reflections, text, and surrounding clutter. - Do not invent architectural elements that are not visible in the source image. - If a building is partially occluded by trees, nearby buildings, street objects, or cropping, infer and complete the missing hidden contour from the visible geometry as faithfully as possible. - Draw continuous clean lines only. Do not output broken, dashed, fragmented, or interrupted segments. - Use thin single black lines only. - White background only. - Pure black lines only. - No fills, no grayscale, no color, no hatch, no shading, and no sketchy rough strokes. - Prefer crisp architectural drafting lines instead of painterly rendering.

Quadrant specification:

- Top-left: Stage 0 / building complex / building. Draw only the outer contour of the main building body or bodies visible in the image. If the image contains multiple main buildings, output multiple separate building outlines. If there is one dominant building, output one building outline. Automatically complete occluded or partially hidden silhouettes from visible geometry, but do not invent extra buildings. Keep only the external envelope with no interior lines at all. No subdivisions, no openings, no structural lines, no decorations, and no secondary contours.

- Top-right: Stage 1 / section. Keep the outer contour exactly unchanged, then add a very small number of simple structural division lines that describe the main sectional organization of the building. Use only sparse vertical or shape-following sectional boundaries. Do not add floor grids, windows, ornament, texture, or fine facade detail.

- Bottom-left: Stage 2 / floor. Keep all previous lines exactly, then add the main floor division lines. Use only sparse horizontal lines that indicate the major floor rhythm and story organization. Do not add windows, panes, ornament, or texture.

- Bottom-right: Stage 3 / windows and doors. Keep all previous lines exactly, then add only the simplest window and door opening shapes. Use sparse single-line geometric outlines only. Adjacent openings may be merged into larger simplified groups when appropriate. Do not add panes, muntins, grills, ornament, or texture.

Critical constraints:

- Use only the original photo as the source of building content.
- Use the example image only as a reference for hierarchy presentation and abstraction style.
- Never leak details from later stages into earlier stages.
- All geometry in a later stage must be derived from and contained within structures already present in earlier stages.
- Do not introduce new building parts, new external masses, or new external elements in later stages.
- Keep each building body as the only subject.
- All stages must remain simple, sparse, and geometric.

### 3. Stage 4-6: refinement prompt template

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#### **Prompt:**

You are an architectural hierarchy renderer for the project.

Goal:

Transform the input building photo into exactly one architectural line drawing stage.

Output format:

- Output one image only.
- Keep the same building viewpoint, framing, scale, perspective, and proportions as the source image.
- No captions, no labels, no numbers, no legends, and no decorative borders.

Global rules:

- Preserve the original building's proportions, silhouette, framing, perspective, and overall geometry.
- Keep only the main building subjects. Remove sky, clouds, trees, ground textures, street furniture, people, cars, shadows, reflections, text, and surrounding clutter.
- Do not invent architectural elements that are not visible in the source image.
- If a building is partially occluded by trees, nearby buildings, street objects, or cropping, infer and complete the missing hidden building contour from the visible geometry as faithfully as possible.
- Draw continuous clean lines. Do not output broken, dashed, fragmented, or interrupted line segments.
- Keep lines as thin and fine as possible. Avoid bold, thick, heavy, or chunky outlines.
- White background only.
- Pure black lines only.
- Absolutely no black fills, no solid black regions, no heavy shadow masses, and no painted black patches.
- Never render windows, doors, shadows, sky, ground, or facade areas as filled black shapes.
- No grayscale, no color, no fills, no hatch, no shading, and no sketchy rough strokes.
- Prefer crisp architectural drafting lines instead of painterly rendering.

Stage target:

- Stage name: **{stage.label}**
- Stage description: **{stage.description}**
- Stage instruction: **{stage.instruction}**

Hierarchy continuity:

- The second reference image is the previous hierarchy stage. Recreate that previous stage faithfully and add only the newly allowed lines for the current stage. Do not remove valid earlier geometry.
- Only include details allowed at this exact stage.
- Never leak details from later stages into earlier stages.

Critical constraint:

- Strictly do not introduce any new building bodies, volumes, or structural parts that were not present in earlier stages. Only refine or subdivide existing geometry.
- If the source image has no visible windows, panes, ornaments, or other details, do not add them.
- Keep each building body as the only subject.

## 4. Stage-specific instructions for Stage 4-6

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### 4.1 Stage 4: structural refinement

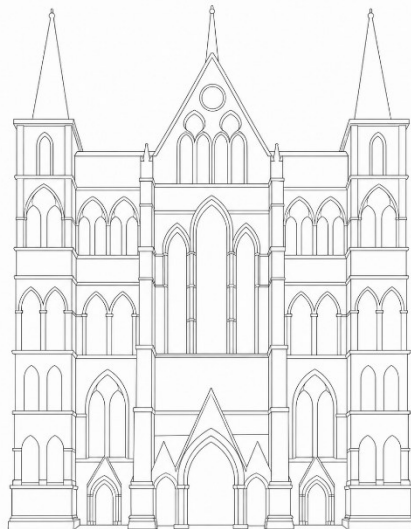


Figure 3: The result of Stage 4: structural refinement

**Description:** add a small number of structural member lines that separate, support, or organize existing openings and masses.

**Prompt:**

Keep all previous lines exactly, then add a small number of simple structural member lines that subdivide and support the existing building parts.

Focus on clear architectural separators and supporting elements, such as:

- columns - pilasters - vertical separators between arches or openings
- simple supports beside doors - drum divisions - major vertical or horizontal member boundaries that organize the facade

Constraints:

- Add only a few new lines overall - Use only thin single black lines
- Keep all additions simple, geometric, and clearly visible in the

source image - Lines should describe structural members that separate, support, or organize existing openings and masses - Prefer full member outlines or clear separator lines over tiny local marks - The result should make the facade read as an organized structural framework

Strictly forbidden:

- No decorative trims, cornices, carvings, reliefs, or ornamental flourishes
- No brick joints, texture fields, or material patterns
- No dense pane grids or tiny repeated interior details
- No shading or filled shapes
- No new outer building parts or new masses

The result should introduce simple structural members such as columns, pilasters, and opening separators, while remaining sparse and non-decorative.

#### 4.2 Stage 5: simple ornament



Figure 4: The result of Stage 5: simple ornament

**Description:** add a small number of simplified decorative or articulating lines along the boundaries of existing members and openings.

#### **Prompt:**

Keep all previous lines exactly, then add a small number of simple decorative or articulating lines along the boundaries of existing architectural members.

Focus on boundary refinement of already existing elements, such as:

- cornice lines;
- arch surrounds;
- window and door surrounds;
- simple column or pilaster edge accents;
- base trims;
- roof-edge profiles;
- simple molding or band lines;
- very simple window subdivision lines inside existing window openings

Constraints:

- Use only thin single black lines;
- Keep all additions simple, clean, and geometric;
- Add detail mainly along the edges of existing members, openings, and major boundaries;
- Window subdivision must be minimal:

only 1 - 4 simple lines per window or window group; - Window subdivision should remain secondary and simplified, not dominant;- Ornament must remain sparse and clearly secondary to structure

Strictly forbidden:

- No new structural subdivision;
- No new columns, pilasters, or opening separators;
- No full pane grids or dense repeated grill patterns;
- No brick joints, texture fields, carvings, sculptures, or dense decorative clutter;
- No shading or filled regions;
- No final full-detail rendering;

The result should refine the boundaries of existing structural members and openings with simple ornament, while adding only very minimal window subdivision and remaining clearly less detailed than the final stage.

#### 4.3 Stage 6: full detail



Figure 5: The result of Stage 6: full detail

**Description:** add the remaining visible facade details, including joints and final trims.

**Prompt:**

Keep all previous lines exactly, then add the remaining visible facade details, brick or block joints, final trims, and the full architectural line representation. If the source image does not show a detail clearly, do not invent it. Still output only black lines on white.